

Midterm 1 Review

ECON 441: Introduction to Mathematical Economics

Div Bhagia

What's on the exam

Modules I and II: Preliminaries and Linear Algebra (Lectures 1–7)

Chiang & Wainwright: 2.2–2.6, p. 163, 4.1–4.7, 5.1–5.5

The help sheet provided with the exam is on the course website.
Everything else you should know.

Work through the sample exam under exam conditions before checking the solutions.

Sets

$\mathbb{R}$ is the set of real numbers (rational and irrational)

$x \in \mathbb{R}$ denotes that x belongs to the set of real numbers

Consider the universe of all real numbers, with set A given by

$$A = \{x | x > 0\}$$

What is the complement of A ?

Set Relations and Operations

Consider the following sets:

$$A = \{x | x > 0\}$$

$$B = \{x | x \text{ is a positive integer}\}$$

$$C = \{x | 1 < x < 5\}$$

- Is $A = B$? Is $B \subset A$?
- Are C and B disjoint sets?
- What is $A \cap B$? What about $A \cup B$?
- What is $B \cap C$?

Subsets

$\emptyset$: empty or null set

What are all possible subsets of

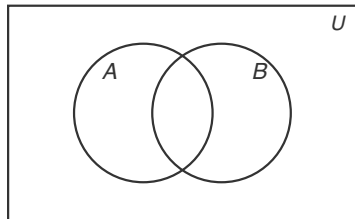
$$S = \{a, b, c\}$$

Always 2^n subsets. Here $n = 3$, so 8 subsets.

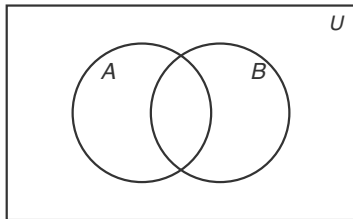
Set Operations: Venn Diagrams

Shade the following regions:

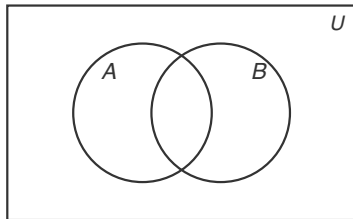
$$A \cup B$$



$$A \cap B$$



$$A^c$$



Cartesian Product

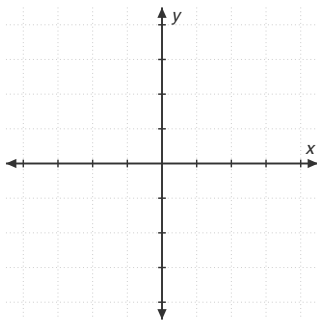
The **Cartesian product** of two sets A and B is the set of all ordered pairs whose first element comes from A and second from B :

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

Example. $A = \{1, 2\}$, $B = \{3, 4\}$. What is $A \times B$?

Cartesian Plane

$$\mathbb{R}^2 = \{(x, y) | x \in \mathbb{R}, y \in \mathbb{R}\}$$



Can have $\mathbb{R}^3, \mathbb{R}^4, \dots, \mathbb{R}^n$

Relations and Functions

Relation: subset of the Cartesian product, e.g. $\{(x, y) | y \leq x\}$

Function: a relation where for each x there is a unique y

$$f : X \rightarrow Y, \quad y = f(x)$$

X : domain, Y : codomain, $f(X)$: range

Is $y = x^2$ a function? Is $\{(x, y) \mid y^2 = x\}$ a function?

Inverse of a Function

Function $y = f(x)$ has an inverse if it is a one-to-one mapping, i.e. each value of y is associated with a unique value of x .

The inverse function

$$x = f^{-1}(y)$$

returns the corresponding value of x for each y .

One-to-one mapping is unique to strictly monotonic functions.

Find the inverse of $y = 3x - 2$. Does $y = x^2$ have an inverse?

Summation Notation

$$\sum_{i=1}^N x_i = x_1 + x_2 + \dots + x_N$$

Things you **can** do:

$$\sum_{i=1}^N bx_i = b \sum_{i=1}^N x_i \qquad \sum_{i=1}^N (a + bx_i) = aN + b \sum_{i=1}^N x_i$$

Things you **cannot** do:

$$\sum_{i=1}^N x_i y_i \neq \sum_{i=1}^N x_i \times \sum_{i=1}^N y_i \qquad \sum_{i=1}^N x_i^a \neq \left(\sum_{i=1}^N x_i \right)^a$$

Necessary vs. Sufficient Conditions

q is a **necessary** condition for p if $p \implies q$

q is a **sufficient** condition for p if $q \implies p$

If both hold, $p \iff q$ and each is **necessary and sufficient** for the other

Example. p : the number is divisible by 4; q : the number is even. Which is it?

Matrices

A matrix with m rows and n columns is an $m \times n$ matrix

$$A = [a_{ij}] \quad i = 1, 2, \dots, m; j = 1, 2, \dots, n$$

Square matrix: equal number of rows and columns

Two matrices are **equal** if all their elements are identical

What is the dimension of $A = \begin{bmatrix} 2 & 3 & 1 \\ -1 & 4 & 6 \end{bmatrix}$?

Addition and Scalar Multiplication

Element-by-element, and the matrices must have the same dimension:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 8 \\ -2 & 3 \end{bmatrix}$$

Find $A + B$, $A - B$, and $2B$.

Matrix Multiplication

Only possible to multiply $A_{m \times n}$ and $B_{p \times q}$ to get AB if $n = p$:

number of columns in A = number of rows in B

$$C = AB, \quad c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik}b_{kj}$$

The element c_{ij} comes from the i th row of A and the j th column of B .

Matrix Multiplication

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & -6 & -2 \end{bmatrix}_{2 \times 3} \quad B = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}_{3 \times 1}$$

Is AB defined? What is its dimension? Find it.

What about BA ?

Vectors and Linear Dependence

Matrices with only one column are **column vectors**; with only one row, **row vectors**

A set of vectors is **linearly dependent** if and only if any one of them can be expressed as a linear combination of the remaining vectors

Are these linearly independent?

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad u_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad u_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Special Matrices

Identity matrix I : 1s on the principal diagonal, 0s elsewhere; acts like 1, so $AI = IA = A$

Null matrix: all elements 0; $A + 0 = A$ and $A0 = 0$

Idempotent: A is idempotent if $A = A^k$, i.e. it is unchanged when multiplied by itself

Is an identity matrix idempotent?

Transpose of a Matrix

A^T : interchange rows and columns

$$(A^T)^T = A \quad (A + B)^T = A^T + B^T \quad (AB)^T = B^T A^T$$

A is **symmetric** if $A^T = A$, **skew-symmetric** if $A^T = -A$, and **orthogonal** if $A^T A = I$

Find A^T for $A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & -6 & 2 \end{bmatrix}$

Inverse of a Matrix

For a **square** matrix A , its inverse A^{-1} is defined as

$$AA^{-1} = A^{-1}A = I$$

If the inverse exists, A is **nonsingular**

$$\left(A^{-1}\right)^{-1} = A \quad (AB)^{-1} = B^{-1}A^{-1} \quad (A^T)^{-1} = \left(A^{-1}\right)^T$$

Conditions for Nonsingularity

Squareness is **necessary** but not **sufficient**

Sufficient condition: **rows (or equivalently) columns are linearly independent**

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Which is singular, and which is nonsingular?

Determinant

The determinant $|A|$ is a unique scalar associated with a **square** matrix A

For a 2×2 matrix:

$$|A| = a_{11}a_{22} - a_{12}a_{21}$$

A **minor** $|M_{ij}|$ deletes the i th row and j th column; the cofactor is

$$|C_{ij}| = (-1)^{i+j} |M_{ij}|$$

Then, expanding along any row or column:

$$|A| = \sum_{j=1}^n a_{ij} |C_{ij}|$$

Find the Determinant

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ -1 & 0 & 0 \end{bmatrix}$$

Expand along the third row. Is A nonsingular?

Criteria for Nonsingularity

The following statements are equivalent:

- $|A| \neq 0$
- Rows (or equivalently columns) of A are independent
- A is nonsingular
- A^{-1} exists
- A unique solution to $Ax = b$ ($x^* = A^{-1}b$) exists

Matrix Inversion

Adjoint of a nonsingular $n \times n$ matrix:

$$\text{adj}A = C^T = \begin{bmatrix} |C_{11}| & |C_{21}| & \cdots & |C_{n1}| \\ |C_{12}| & |C_{22}| & \cdots & |C_{n2}| \\ \vdots & \vdots & \cdots & \vdots \\ |C_{1n}| & |C_{2n}| & \cdots & |C_{nn}| \end{bmatrix}$$

The **inverse** of A is defined as:

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

Find the Inverse

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$$

Then try

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ -1 & 0 & 0 \end{bmatrix}$$

Solving a System of Equations

$$Ax = b \quad \implies \quad x^* = A^{-1}b$$

Cramer's rule: form A_k by replacing the k th column of A with b ; then

$$x_k^* = \frac{|A_k|}{|A|}$$

If A is singular, a unique solution does not exist.

A Simple Economic Model

$$q + 2p = 100$$

$$q - 3p = 20$$

Write this as $Ax = b$. What are A , x , and b ?

Solve it two ways: by matrix inversion, and by Cramer's rule.

Few last words

The sample exam and help sheet are on the course website

Practice problems 1–7 and their solutions are the best preparation

Additional office hours this week, or by appointment

Good luck for the exam!